14) Using RAW Sockets to implement packet Sniffing

Aim:

To implement a code using RAW Sockets to implement packet Sniffing.

procedure:

coding:

```
from Scapy.all import Sniff.
from Scapy.layers.inet import Ip, Tcp, UDP,
ICMP.
def packet_callback(packet):
    if IP in packet:
        ip_layer = packet[IP].
        protocol = ip_layer.proto
        Src_ip = ip_layer.src
        def_ip = Ip_layer.dst.
        protocol_name = ""
        if protocol == 1:
            protocol_name = "ICMP"
        elif protocol == 6
            protocol_name = "Tcp"
```

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```
elif protocol == 17:
    protocol_name = "UDP"
else:
    protocol_name = "unknown protocol"
print(f"protocol: {protocol_name}")
print(f"Source Ip: {src_ip}")
print(f"Destination Ip: {dst_ip}")
print("-" * 50)
Sniff(iface = "wi-Fi", prn = packet_callback,
      filter = "ip", store = 0)
```

Output:

```
C:\Users\tkala\Downloads>python packet.py
protocol: TCP
Source Ip: 57.155.141.117
Destination Ip: 192.168.1.6
- - - - - - - - - - - - - - - - - - - - - -
protocol: TCP
Source Ip: 192.168.1.6
Destination Ip: 57.155.141.117
```

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protocol : Tcp

Source Ip : 192.168.1.6

Destination Ip : 57.155.141.117

protocol : Tcp

Source Ip : 192.168.1.6

Destination Ip : 57.155.141.117

protocol : Tcp

Source Ip : 57.155.141.117

Destinat Ip : 192.168.1.6

protocol : Tcp

Source Ip : 57.155.141.117

Destinatio Ip : 192.168.1.6

protocol : UDp

Source Ip : 192.168.1.6

Destinatio Ip : 192.168.1.1

protocol : UDp

source Ip : 192.168.1.6

Destinatio Ip : 192.168.1.1

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protocol: udp

Source Ip : 192.168.1.1

Destination Ip : 192.168.1.1

protocol : udp

Source Ip : 192.168.1.1

Destination : 192.168.1.6

Result:

Therefore, the RAW Sockets to implement packet sniffer executed Successfully.

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